

Week 5

Healthcare Data Cleaning & Data Modeling

Power Query Data Cleaning, Star-Schema Modeling and DAX Measures in Power BI

Name: Kamali V

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1. Task Overview:

1.1 Objectives:

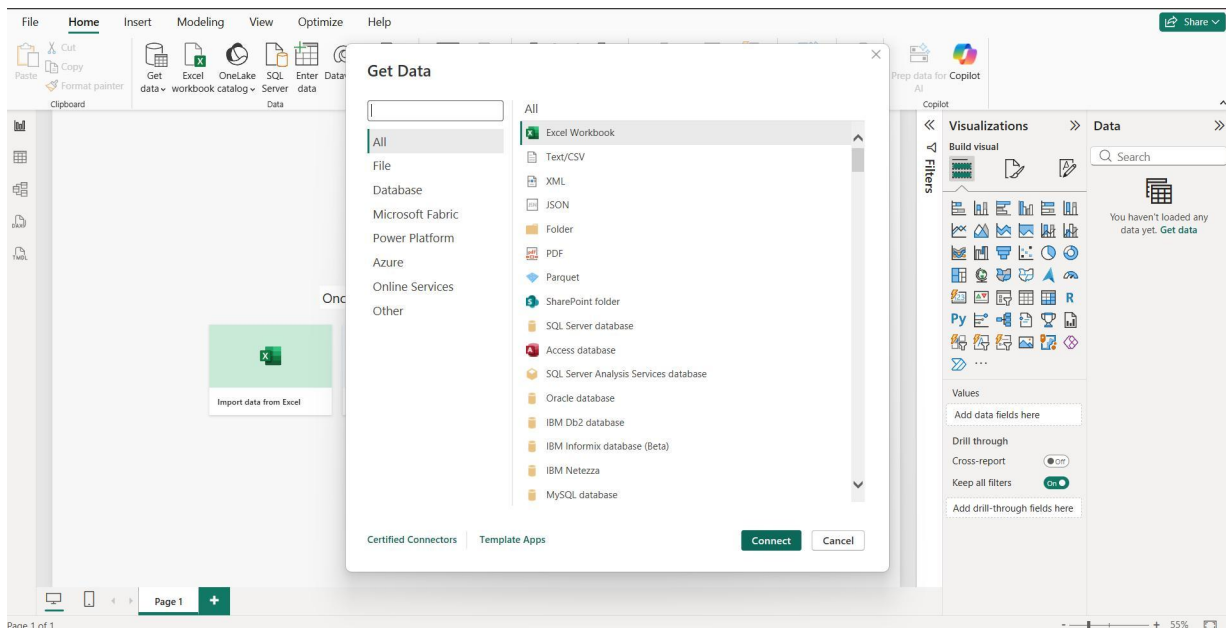
- Import and inspect patient records, admission types, and billing fields in Power Query.
- Handle null values in diagnostic codes and standardize categorical text attributes.
- Create derived conditional columns (e.g., Age Group buckets: 0–18, 19–35, 36–60, 60+).
- Structure dimension tables (Dim_Patient, Dim_Admission) and a billing fact table.
- Create a Measures Table with DAX measures (Total Patients, Total Billing Amount, Avg Billing Amount, Avg Length of Stay, Readmission Rate).

1.2 Tasks & Deliverables:

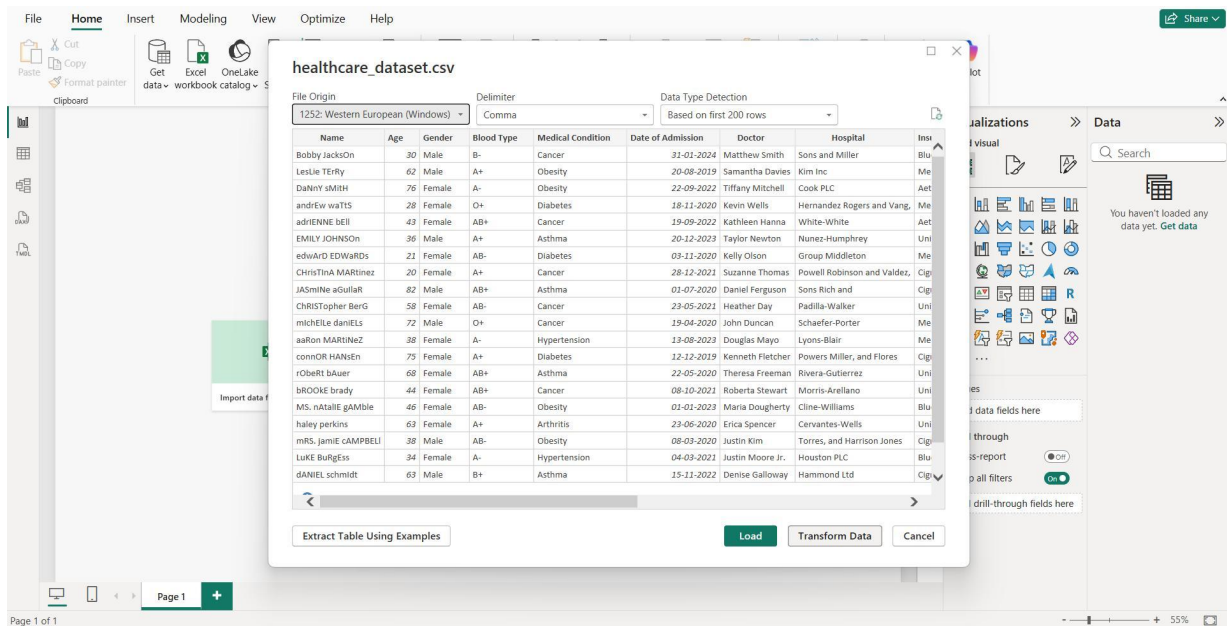
- Group patient ages into logical demographic categories via Power Query.
- Standardize admission types (Emergency, Elective, Urgent).
- Set up a star-schema model with patient and admission dimensions around a billing fact table.
- Add a Measures Table with the core KPI measures, including a Readmission Rate measure.
- Output a cleaned data model ready for clinical analytics.

2. Importing the Dataset:

The healthcare_dataset.csv file was brought into Power BI Desktop through Get Data, choosing the Text/CSV connector from the list of available sources.

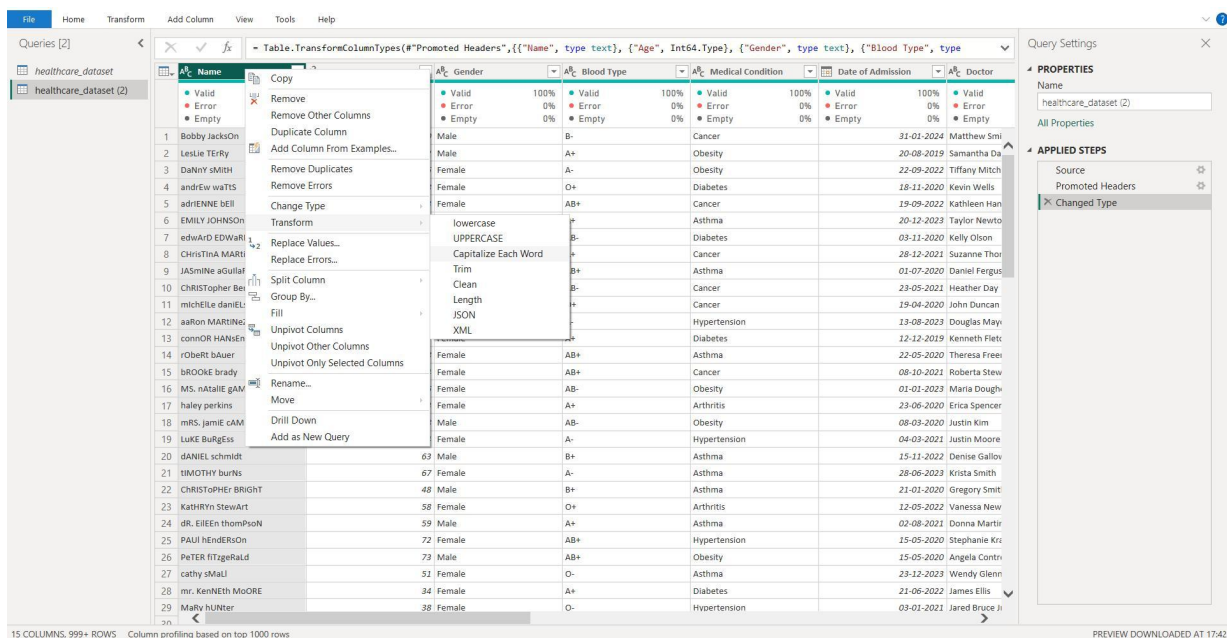


The preview window confirmed the file origin (1252: Western European), the comma delimiter, and data type detection based on the first 200 rows. The preview also showed inconsistent capitalization in the Name column (for example "Bobby JacksOn" and "LesLie TERry"), so Transform Data was chosen to open the Power Query Editor instead of loading the file directly.



3. Standardizing Text Attributes:

Inside Power Query, the data types of the columns were reviewed and set. To standardize the text, the Name column was selected and Transform > Capitalize Each Word was applied from the right-click menu, so every patient name follows one consistent format (for example "Bobby Jackson").



4. Creating the Age Group Conditional Column:

To group patients into demographic categories, Add Column > Conditional Column was opened and the new column was named Age Group.

15 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows. PREVIEW DOWNLOADED AT 19:15

Three rules were defined on the Age column: Age ≤ 18 gives Child, Age ≤ 35 gives Young, and Age ≤ 60 gives Adult. Any age above 60 falls into the Else clause and is labelled Senior.

15 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows. PREVIEW DOWNLOADED AT 19:15

The resulting Age Group column was added as a new step in the Applied Steps pane. The formula bar shows the generated if-then-else expression, and each patient row now carries a Child, Young, Adult, or Senior label.

FileHomeTransformAdd ColumnViewToolsHelp

Queries [1]

fx

= Table.AddColumn("Changed Type", "Age Group", each if [Age] <= 18 then "Child" else if [Age] <= 35 then "Young" else if [Age] <= 60 then

healthcare_dataset

100%Valid0%Error0%Empty

100%Valid0%Error0%Empty

100%Valid0%Error0%Empty

100%Valid0%Error0%Empty

100%Valid0%Error0%Empty

100%Valid0%Error0%Empty

100%Valid0%Error0%Empty

1

18856.28131

328

Urgent

02-02-2024

Paracetamol

Normal

Young

2

33643.32729

265

Emergency

26-08-2019

Ibuprofen

Inconclusive

Senior

3

27955.09608

205

Emergency

07-10-2022

Aspirin

Normal

Senior

4

37909.78241

450

Elective

18-12-2020

Ibuprofen

Abnormal

Young

5

14238.31781

458

Urgent

09-10-2022

Penicillin

Abnormal

Adult

6

48145.11095

389

Urgent

24-12-2023

Ibuprofen

Normal

Adult

7

19580.87234

389

Emergency

15-11-2020

Paracetamol

Inconclusive

Young

8

45820.46272

277

Emergency

07-01-2022

Paracetamol

Inconclusive

Young

9

50119.22279

316

Elective

14-07-2020

Aspirin

Abnormal

Senior

10

19784.63106

249

Elective

22-06-2021

Paracetamol

Inconclusive

Adult

11

12576.79561

394

Urgent

22-04-2020

Paracetamol

Normal

Senior

12

7999.58688

288

Urgent

05-09-2023

Lipitor

Inconclusive

Adult

13

43282.28336

134

Emergency

28-12-2019

Penicillin

Abnormal

Senior

14

33207.70663

309

Urgent

19-06-2020

Lipitor

Normal

Senior

15

40701.59923

182

Urgent

13-10-2021

Paracetamol

Normal

Adult

16

12263.35743

465

Elective

11-01-2023

Aspirin

Inconclusive

Adult

17

24499.8479

114

Elective

14-07-2020

Paracetamol

Normal

Senior

18

17440.46544

449

Urgent

02-04-2020

Paracetamol

Abnormal

Adult

19

18843.02302

260

Elective

14-03-2021

Aspirin

Abnormal

Young

20

23762.20358

465

Elective

22-11-2022

Penicillin

Normal

Senior

21

42.51458855

115

Elective

02-07-2023

Aspirin

Normal

Senior

22

17695.91162

295

Urgent

09-02-2020

Lipitor

Normal

Adult

23

5998.102908

327

Urgent

10-06-2022

Lipitor

Inconclusive

Adult

24

25250.05243

119

Urgent

12-08-2021

Lipitor

Inconclusive

Adult

25

33211.29542

109

Emergency

08-06-2020

Paracetamol

Inconclusive

Senior

26

19746.83201

162

Urgent

20-05-2020

Aspirin

Abnormal

Senior

27

26786.52956

401

Elective

19-01-2024

Ibuprofen

Normal

Adult

28

18834.80134

157

Emergency

30-06-2022

Lipitor

Abnormal

Young

29

32643.29935

223

Emergency

16-01-2021

Penicillin

Normal

Adult

16 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 19:35

5. Adding Surrogate Keys:

To link the tables later, two surrogate keys were created. Using Add Column > Index Column > From 1, an index column starting at 1 was added to give every row a unique number.

FileHomeTransformAdd ColumnViewToolsHelp

Column From Custom Examples * Column

Conditional Column

Index Column

From 1

From Text

From Number

From Date & Time

Custom

Create a new column with an index starting at 1.

0%Valid0%Error0%Empty

0%Valid0%Error0%Empty

0%Valid0%Error0%Empty

0%Valid0%Error0%Empty

0%Valid0%Error0%Empty

0%Valid0%Error0%Empty

0%Valid0%Error0%Empty

1

Bobby

30

Male

B-

Cancer

31-01-2024

Matthew Smit

2

Leslie Terry

62

Male

A+

Obesity

20-08-2019

Samantha Da

3

Danny Smith

76

Female

A-

Obesity

22-09-2022

Tiffany Mitch

4

Andrew Watts

28

Female

O+

Diabetes

18-11-2020

Kevin Wells

5

Adrienne Bell

43

Female

AB+

Cancer

19-09-2022

Kathleen Han

6

Emily Johnson

36

Male

A+

Asthma

20-12-2023

Taylor Newto

7

Edward Edwards

21

Female

AB-

Diabetes

03-11-2020

Kelly Olson

8

Christina Martinez

20

Female

A+

Cancer

28-12-2021

Suzanne Thor

9

Jasmine Aguilar

82

Male

AB+

Asthma

01-07-2020

Daniel Fergus

10

Christopher Berg

58

Female

AB-

Cancer

23-05-2021

Heather Day

11

Michelle Daniels

72

Male

O+

Cancer

19-04-2020

John Duncan

12

Aaron Martinez

38

Female

A-

Hypertension

13-08-2023

Douglas May

13

Connor Hansen

75

Female

A+

Diabetes

12-12-2019

Kenneth Flet

14

Robert Bauer

64

Female

AB+

Asthma

22-05-2020

Theresa Free

15

Brooke Brady

44

Female

AB+

Cancer

08-10-2021

Roberta Stew

16

Ms. Natalie Gamble

46

Female

AB-

Obesity

01-01-2023

Maria Dough

17

Haley Perkins

63

Female

A+

Arthritis

23-06-2020

Erica Spencer

18

Mrs. Jamie Campbell

38

Male

AB-

Obesity

08-03-2020

Justin Kim

19

Luke Burgess

34

Female

A-

Hypertension

04-03-2021

Justin Moore

20

Daniel Schmidt

63

Male

B+

Asthma

15-11-2022

Denise Gallov

21

Timothy Burns

67

Female

A-

Asthma

28-06-2023

Krista Smith

22

Christopher Bright

48

Male

B+

Asthma

21-01-2020

Gregory Smit

23

Kathryn Stewart

58

Female

O+

Arthritis

12-05-2022

Vanessa New

24

Dr. Eileen Thompson

59

Male

A+

Asthma

02-08-2021

Donna Martir

25

Paul Henderson

72

Female

AB+

Hypertension

15-05-2020

Stephanie Kr

26

Peter Fitzgerald

73

Male

AB+

Obesity

15-05-2020

Angela Contr

27

Cathy Small

51

Female

O-

Asthma

23-12-2023

Wendy Glenn

28

Mr. Kenneth Moore

34

Female

A+

Diabetes

21-06-2022

James Ellis

29

Mary Hunter

38

Female

O-

Hypertension

03-02-2021

Jared Bruce Ji

15 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 18:31

A second index column was added the same way. Each new index column was then renamed through the right-click Rename option.

The screenshot shows a data table with columns: Admission Type, Discharge Date, Medication, Test Results, and Index. A right-click context menu is open over the 'Index' column header, showing options like Copy, Remove, Duplicate Column, Add Column From Examples..., Remove Duplicates, Remove Errors, Change Type, Transform, Replace Values..., Replace Errors..., Group By..., Fill, Unpivot Columns, Unpivot Other Columns, Unpivot Only Selected Columns, Rename..., Move, Drill Down, and Add as New Query. The 'Rename...' option is highlighted. The table has 17 columns and 999+ rows. The status bar at the bottom indicates '17 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

The two index columns were renamed Patient ID and Admission ID, so that the same keys can be carried into the patient, admission, and billing tables.

The screenshot shows the same data table, but the 'Index' column has been renamed to 'Admission ID'. The right-click context menu is no longer open. The table now has columns: Admission Type, Discharge Date, Medication, Test Results, Patient ID, and Admission ID. The status bar at the bottom indicates '17 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

With the Age Group column and both keys in place, the main healthcare_dataset query now holds 18 columns and serves as the source for the dimension and billing queries.

FileHomeTransformAdd ColumnViewToolsHelp

Queries [4]

healthcare_dataset

Dim_Patient

Billing

Dim_Admission

Table.RenameColumns(#{Added Index1},{('Index', 'Admission ID'), ('Index.1', 'Patient ID')})

Name

Age

Gender

Blood Type

Medical Condition

Date of Admission

Doctor

Valid

100%

Valid

100%

Valid

100%

Valid

100%

Valid

100%

Valid

100%

Error

0%

Error

0%

Error

0%

Error

0%

Error

0%

Error

0%

Empty

0%

Empty

0%

Empty

0%

Empty

0%

Empty

0%

Empty

0%

1

Bobby Jackson

30

Male

B-

Cancer

31-01-2024

Matthew Sml

2

Leslie Terry

62

Male

A+

Obesity

20-08-2019

Samantha Da

3

Danny Smith

76

Female

A-

Obesity

22-09-2022

Tiffany Mitch

4

Andrew Watts

28

Female

O+

Diabetes

28-11-2020

Kevin Wells

5

Adrienne Bell

43

Female

AB+

Cancer

29-09-2022

Kathleen Han

6

Emily Johnson

36

Male

A+

Asthma

20-12-2023

Taylor Newto

7

Edward Edwards

21

Female

AB-

Diabetes

03-12-2020

Kelly Olson

8

Christina Martinez

20

Female

A+

Cancer

28-12-2021

Suzanne Thor

9

Jasmine Aguilar

82

Male

AB+

Asthma

01-07-2020

Daniel Fergus

10

Christopher Berg

58

Female

AB-

Cancer

23-05-2021

Heather Day

11

Michelle Daniels

72

Male

O+

Cancer

19-04-2020

John Duncan

12

Aaron Martinez

38

Female

A-

Hypertension

13-08-2023

Douglas May

13

Connor Hansen

75

Female

A+

Diabetes

12-12-2019

Kenneth Fletc

14

Robert Bauer

68

Female

AB+

Asthma

22-05-2020

Theresa Freee

15

Brooke Brady

44

Female

AB+

Cancer

08-10-2021

Roberta Stew

16

Ms. Natalie Gamble

46

Female

AB-

Obesity

01-01-2023

Maria Dough

17

Haley Perkins

63

Female

A+

Arthritis

23-06-2020

Erica Spencer

18

Mrs. Jamie Campbell

38

Male

AB-

Obesity

08-09-2020

Justin Kim

19

Luke Burgess

34

Female

A-

Hypertension

04-03-2021

Justin Moore

20

Daniel Schmidt

63

Male

B+

Asthma

15-11-2022

Denise Gallov

21

Timothy Burns

67

Female

A-

Asthma

28-08-2023

Krista Smith

22

Christopher Bright

48

Male

B+

Asthma

21-01-2020

Gregory Smit

23

Kathryn Stewart

58

Female

O+

Arthritis

12-05-2022

Vanessa New

24

Dr. Eileen Thompson

59

Male

A+

Asthma

02-08-2021

Donna Martir

25

Paul Henderson

72

Female

AB+

Hypertension

15-05-2020

Stephanie Kri

26

Peter Fitzgerald

73

Male

AB+

Obesity

15-05-2020

Angela Contr

27

Cathy Small

51

Female

O-

Asthma

23-12-2023

Wendy Glenn

28

Mr. Kenneth Moore

34

Female

A+

Diabetes

21-06-2022

James Ellis

29

Mary Hunter

38

Female

O-

Hypertension

03-01-2021

Jared Bruce Ji

30

18 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 19:41

6. Building the Dim_Patient Dimension Table:

A new query, Dim_Patient, was created by referencing healthcare_dataset. The columns that do not describe the patient (such as Medical Condition) were removed with the right-click Remove option.

FileHomeTransformAdd ColumnViewToolsHelp

Queries [2]

healthcare_dataset

Dim_Patient

healthcare_dataset

Table.RenameColumns(#{Added Index1},{('Index', 'Admission ID'), ('Index.1', 'Patient ID')})

Name

Age

Gender

Blood Type

Medical Condition

Date of Admission

Doctor

Valid

100%

Valid

100%

Valid

100%

Valid

100%

Valid

100%

Valid

100%

Error

0%

Error

0%

Error

0%

Error

0%

Error

0%

Error

0%

Empty

0%

Empty

0%

Empty

0%

Empty

0%

Empty

0%

Empty

0%

1

Bobby Jackson

30

Male

B-

Cancer

31-01-2024

Matthew Sml

2

Leslie Terry

62

Male

A+

Obesity

20-08-2019

Samantha Da

3

Danny Smith

76

Female

A-

Obesity

22-09-2022

Tiffany Mitch

4

Andrew Watts

28

Female

O+

Diabetes

28-11-2020

Kevin Wells

5

Adrienne Bell

43

Female

AB+

Cancer

29-09-2022

Kathleen Han

6

Emily Johnson

36

Male

A+

Asthma

20-12-2023

Taylor Newto

7

Edward Edwards

21

Female

AB-

Diabetes

03-12-2020

Kelly Olson

8

Christina Martinez

20

Female

A+

Cancer

28-12-2021

Suzanne Thor

9

Jasmine Aguilar

82

Male

AB+

Asthma

01-07-2020

Daniel Fergus

10

Christopher Berg

58

Female

AB-

Cancer

23-05-2021

Heather Day

11

Michelle Daniels

72

Male

O+

Cancer

19-04-2020

John Duncan

12

Aaron Martinez

38

Female

A-

Hypertension

13-08-2023

Douglas May

13

Connor Hansen

75

Female

A+

Diabetes

12-12-2019

Kenneth Fletc

14

Robert Bauer

68

Female

AB+

Asthma

22-05-2020

Theresa Freee

15

Brooke Brady

44

Female

AB+

Cancer

08-10-2021

Roberta Stew

16

Ms. Natalie Gamble

46

Female

AB-

Obesity

01-01-2023

Maria Dough

17

Haley Perkins

63

Female

A+

Arthritis

23-06-2020

Erica Spencer

18

Mrs. Jamie Campbell

38

Male

AB-

Obesity

08-09-2020

Justin Kim

19

Luke Burgess

34

Female

A-

Hypertension

04-03-2021

Justin Moore

20

Daniel Schmidt

63

Male

B+

Asthma

15-11-2022

Denise Gallov

21

Timothy Burns

67

Female

A-

Asthma

28-08-2023

Krista Smith

22

Christopher Bright

48

Male

B+

Asthma

21-01-2020

Gregory Smit

23

Kathryn Stewart

58

Female

O+

Arthritis

12-05-2022

Vanessa New

24

Dr. Eileen Thompson

59

Male

A+

Asthma

02-08-2021

Donna Martir

25

Paul Henderson

72

Female

AB+

Hypertension

15-05-2020

Stephanie Kri

26

Peter Fitzgerald

73

Male

AB+

Obesity

15-05-2020

Angela Contr

27

Cathy Small

51

Female

O-

Asthma

23-12-2023

Wendy Glenn

28

Mr. Kenneth Moore

34

Female

A+

Diabetes

21-06-2022

James Ellis

29

Mary Hunter

38

Female

O-

Hypertension

03-01-2021

Jared Bruce Ji

30

16 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 19:39

After the Removed Columns step, Dim_Patient keeps six columns: Name, Age, Gender, Blood Type, Age Group, and Patient ID.

FileHomeTransformAdd ColumnViewToolsHelp

Queries [2]

healthcare_dataset

Dim_Patient

Table.RemoveColumns(Source,["Medical Condition", "Date of Admission", "Doctor", "Hospital", "Insurance Provider", "Billing Amount",

Name

Age

Gender

Blood Type

Age Group

Patient ID

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

1

Bobby Jackson

30

Male

B-

Young

1

2

Leslie Terry

62

Male

A+

Senior

2

3

Danny Smith

76

Female

A-

Senior

3

4

Andrew Watts

28

Female

O+

Young

4

5

Adrienne Bell

43

Female

AB+

Adult

5

6

Emily Johnson

36

Male

A+

Adult

6

7

Edward Edwards

21

Female

AB-

Young

7

8

Christina Martinez

20

Female

A+

Young

8

9

Jasmine Aguilar

82

Male

AB+

Senior

9

10

Christopher Berg

58

Female

AB-

Adult

10

11

Michelle Daniels

72

Male

O+

Senior

11

12

Aaron Martinez

38

Female

A-

Adult

12

13

Connor Hansen

75

Female

A+

Senior

13

14

Robert Bauer

68

Female

AB+

Senior

14

15

Brooke Brady

44

Female

AB+

Adult

15

16

Ms. Natalie Gamble

46

Female

AB-

Adult

16

17

Hailey Perkins

63

Female

A+

Senior

17

18

Mrs. Jamie Campbell

38

Male

AB-

Adult

18

19

Luke Burgess

34

Female

A-

Young

19

20

Daniel Schmidt

64

Male

B+

Senior

20

21

Timothy Burns

67

Female

A-

Senior

21

22

Christopher Bright

48

Male

B+

Adult

22

23

Kathryn Stewart

58

Female

O+

Adult

23

24

Dr. Eileen Thompson

59

Female

A+

Adult

24

25

Paul Henderson

72

Female

AB+

Senior

25

26

Peter Fitzgerald

73

Male

AB+

Senior

26

27

Cathy Small

51

Female

O-

Adult

27

28

Mr. Kenneth Moore

34

Female

A+

Young

28

29

Mary Hunter

38

Female

O-

Adult

29

30

Isabell Oliver

43

Female

AB-

Adult

30

6 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 1943

7. Building the Billing Table:

The Billing query was created from the main query by removing all patient and admission descriptor columns. It keeps Date of Admission, Insurance Provider, Billing Amount, Discharge Date, plus the Admission ID and Patient ID keys, so each billing record can be traced back to one patient and one admission.

FileHomeTransformAdd ColumnViewToolsHelp

Queries [3]

healthcare_dataset

Dim_Patient

Billing

Table.RemoveColumns(Source,["Name", "Age", "Gender", "Blood Type", "Doctor", "Hospital", "Room Number", "Medical Condition", "Admission

Date of Admission

Insurance Provider

Billing Amount

Discharge Date

Admission ID

Patient ID

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

1

31-01-2024

Blue Cross

18856.28131

02-02-2024

1

1

2

20-08-2019

Medicare

33643.32729

26-08-2019

2

2

3

22-09-2022

Aetna

27955.09608

07-10-2022

3

3

4

18-11-2020

Medicare

37909.78241

18-12-2020

4

4

5

19-09-2022

Aetna

14238.31781

09-10-2022

5

5

6

20-12-2023

UnitedHealthcare

48145.11095

24-12-2023

6

6

7

03-11-2020

Medicare

19580.87234

15-11-2020

7

7

8

28-12-2021

Cigna

45820.46272

07-01-2022

8

8

9

01-07-2020

Cigna

50119.22279

14-07-2020

9

9

10

23-05-2021

UnitedHealthcare

19784.63106

22-06-2021

10

10

11

19-04-2020

Medicare

12576.79561

22-04-2020

11

11

12

13-08-2023

Medicare

7999.58688

05-09-2023

12

12

13

12-12-2019

Cigna

43282.28336

28-12-2019

13

13

14

22-05-2020

UnitedHealthcare

33207.70663

19-06-2020

14

14

15

08-10-2021

UnitedHealthcare

40701.59923

13-10-2021

15

15

16

01-01-2023

Blue Cross

12263.35743

11-01-2023

16

16

17

23-06-2020

UnitedHealthcare

24499.8479

14-07-2020

17

17

18

08-03-2020

Cigna

17440.46544

02-04-2020

18

18

19

04-03-2021

Blue Cross

18843.02302

14-03-2021

19

19

20

15-11-2022

Cigna

23762.20358

22-11-2022

20

20

21

28-06-2023

Blue Cross

42.51458855

02-07-2023

21

21

22

21-01-2020

Aetna

17695.91162

09-02-2020

22

22

23

12-05-2022

Aetna

5998.102908

10-06-2022

23

23

24

02-08-2021

Aetna

25250.05243

12-08-2021

24

24

25

15-05-2020

Medicare

33211.29542

08-06-2020

25

25

26

15-05-2020

Medicare

19746.83201

20-05-2020

26

26

27

23-12-2022

Blue Cross

26786.52956

19-01-2024

27

27

28

21-06-2022

UnitedHealthcare

18834.80134

30-06-2022

28

28

29

03-01-2021

Cigna

32643.29935

16-01-2021

29

29

30

02-10-2022

Aetna

67673.013064

13-10-2022

30

30

6 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

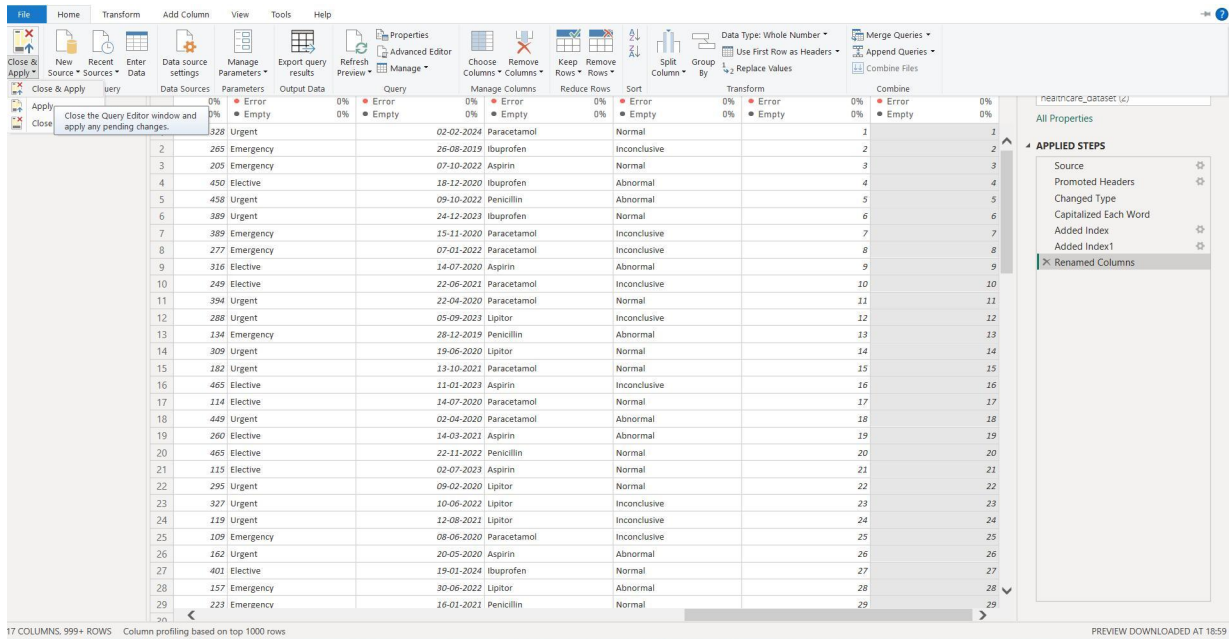
PREVIEW DOWNLOADED AT 1943

8. Building the Dim_Admission Dimension Table:

In the same way, a Dim_Admission query was created from the main query, keeping the admission-related attributes: Admission ID, Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number, and Test Results. Admission Type values are consistent across the data (Emergency, Elective, and Urgent), which meets the standardization task. Admission ID acts as the primary key of this table.

9. Applying Changes and Loading the Model:

Once all queries were ready, Close & Apply was used to close the Query Editor and load the cleaned tables into Power BI Desktop.



Patient ID	Urgency	Date	Medication	Status
328	Urgent	02-02-2024	Paracetamol	Normal
265	Emergency	26-08-2019	Ibuprofen	Inconclusive
205	Emergency	07-10-2022	Aspirin	Normal
450	Elective	18-12-2020	Ibuprofen	Abnormal
458	Urgent	09-10-2022	Penicillin	Abnormal
389	Urgent	24-12-2023	Ibuprofen	Normal
389	Emergency	15-11-2020	Paracetamol	Inconclusive
277	Emergency	07-01-2022	Paracetamol	Inconclusive
316	Elective	14-07-2020	Aspirin	Abnormal
249	Elective	22-06-2021	Paracetamol	Inconclusive
394	Urgent	22-04-2020	Paracetamol	Normal
288	Urgent	05-09-2023	Lipitor	Inconclusive
134	Emergency	28-12-2019	Penicillin	Abnormal
309	Urgent	19-06-2020	Lipitor	Normal
182	Urgent	13-10-2021	Paracetamol	Normal
465	Elective	11-01-2023	Aspirin	Inconclusive
114	Elective	14-07-2020	Paracetamol	Normal
449	Urgent	02-04-2020	Paracetamol	Abnormal
260	Elective	14-03-2021	Aspirin	Abnormal
465	Elective	22-11-2022	Penicillin	Normal
115	Elective	02-07-2023	Aspirin	Normal
295	Urgent	09-02-2020	Lipitor	Normal
327	Urgent	10-06-2022	Lipitor	Inconclusive
119	Urgent	12-08-2021	Lipitor	Inconclusive
109	Emergency	08-06-2020	Paracetamol	Inconclusive
162	Urgent	20-05-2020	Aspirin	Abnormal
401	Elective	19-01-2024	Ibuprofen	Normal
157	Emergency	30-06-2022	Lipitor	Abnormal
223	Emergency	16-01-2021	Penicillin	Normal

10. Final Data Model:

The Model view shows the loaded tables: Dim_Patient, Dim_Admission, Billing, and the original healthcare_dataset query. Billing holds Patient ID and Admission ID, matching the primary keys of the two dimension tables, so it can act as the central fact table of the star schema.

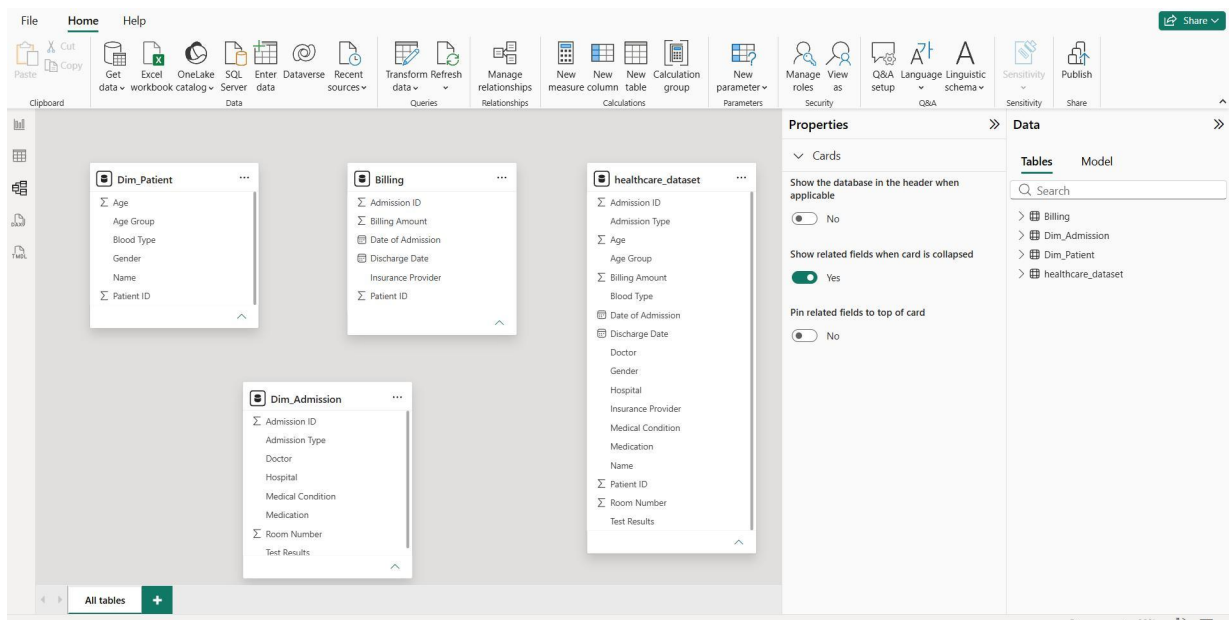
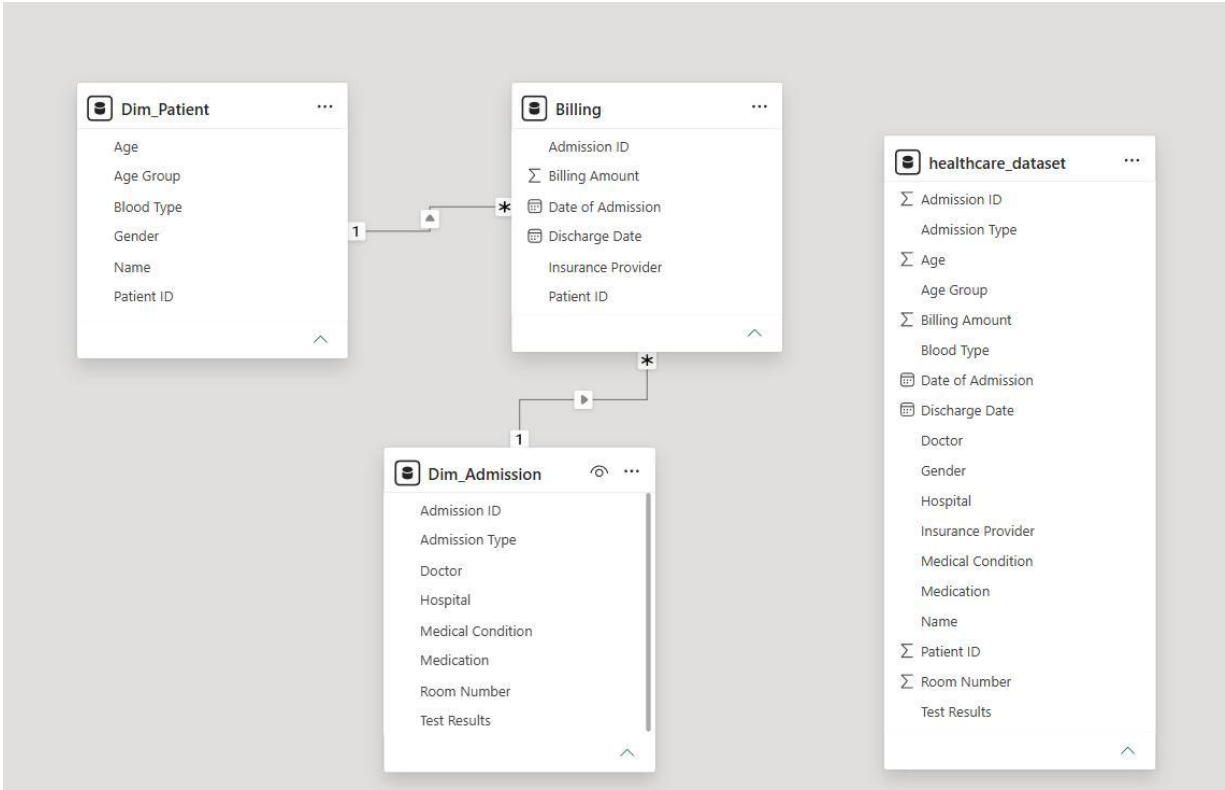


Table	Fields
Dim_Patient	Age, Age Group, Blood Type, Gender, Name, Patient ID
Dim_Admission	Admission ID, Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number, Test Results
Billing	Admission ID, Billing Amount, Date of Admission, Discharge Date, Insurance Provider, Patient ID
healthcare_dataset	Admission ID, Admission Type, Age, Age Group, Billing Amount, Blood Type, Date of Admission, Discharge Date, Doctor, Gender, Hospital, Insurance Provider, Medical Condition, Medication, Name, Patient ID, Room Number, Test Results

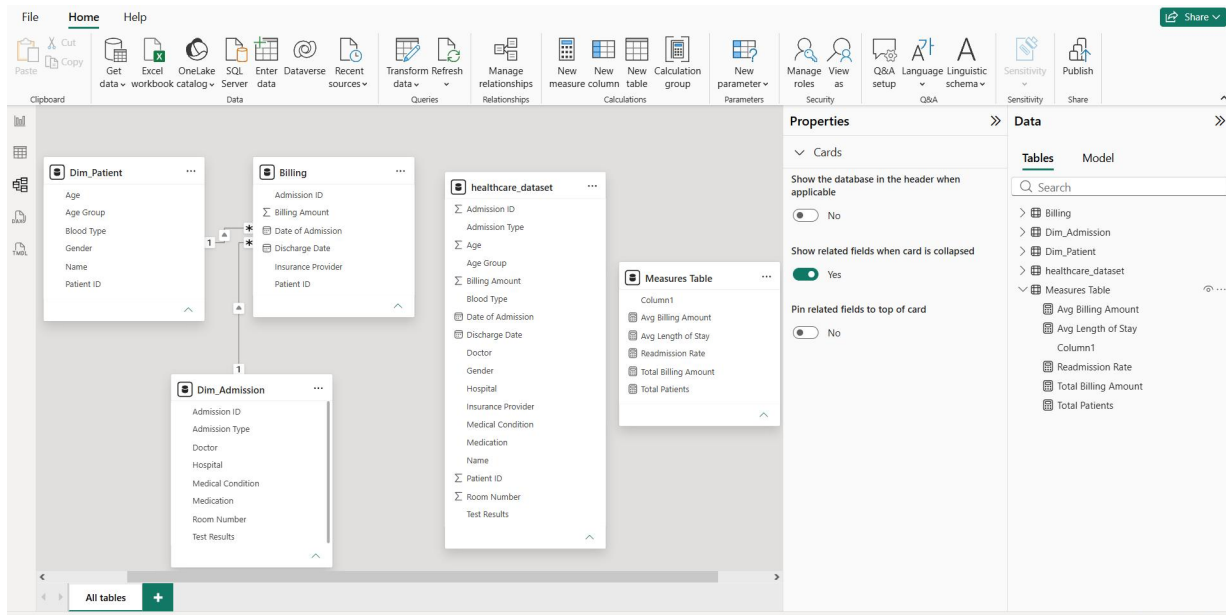
Relationships were then defined in the model. Dim_Patient is linked to Billing on Patient ID and Dim_Admission is linked to Billing on Admission ID. Both are one-to-many relationships (1 on the dimension side, * on the Billing side) with single-direction filtering from the dimension to the fact table, giving the star schema with Billing at the centre.



11. Creating the Measures Table:

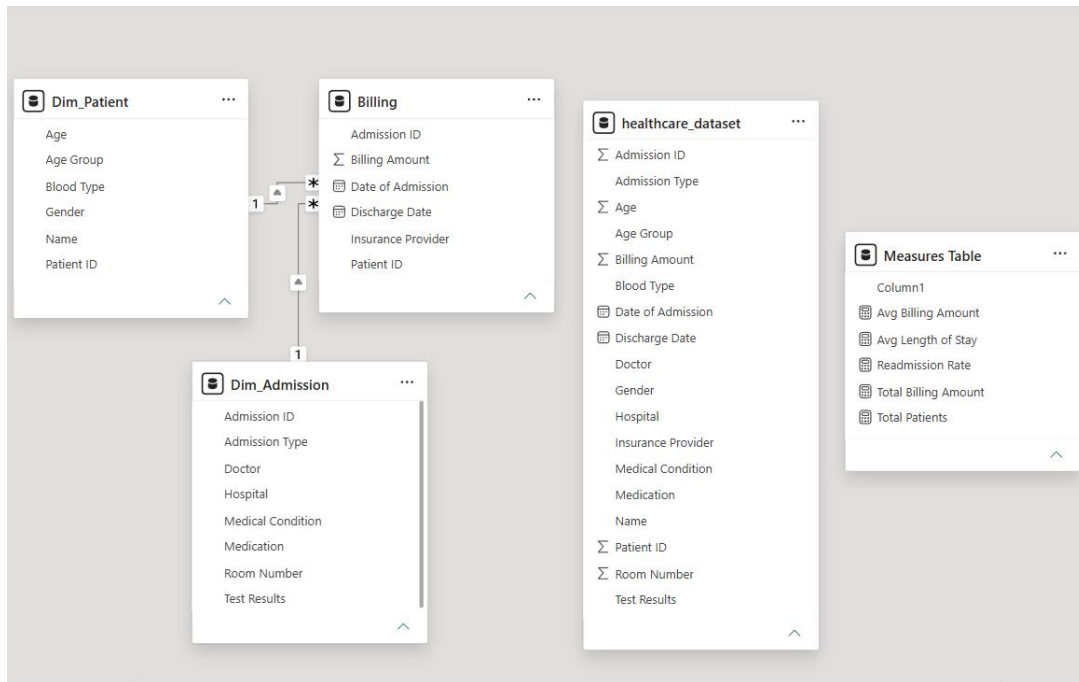
To keep the calculations organised, a separate table named Measures Table was added to the model to hold all DAX measures instead of scattering them across the data tables. Five measures were created in it: Total Patients, Total Billing Amount, Avg Billing Amount, Avg Length of Stay, and Readmission Rate. At this stage the table still shows a placeholder column, Column1, alongside the measures.

The screenshot below shows the Model view with the Measures Table card next to the star schema. The Data pane on the right lists the Measures Table with its five measures, together with Billing, Dim_Admission, Dim_Patient, and healthcare_dataset.



12. Measures Table in the Data Model:

A closer view of the Model confirms the Measures Table sits beside Dim_Patient, Billing, Dim_Admission, and healthcare_dataset. Because it only stores measures, it is not connected to any other table, and the one-to-many relationships from Dim_Patient and Dim_Admission to Billing stay unchanged. The measure icons on its card show that Avg Billing Amount, Avg Length of Stay, Readmission Rate, Total Billing Amount, and Total Patients are calculated measures and not ordinary columns.



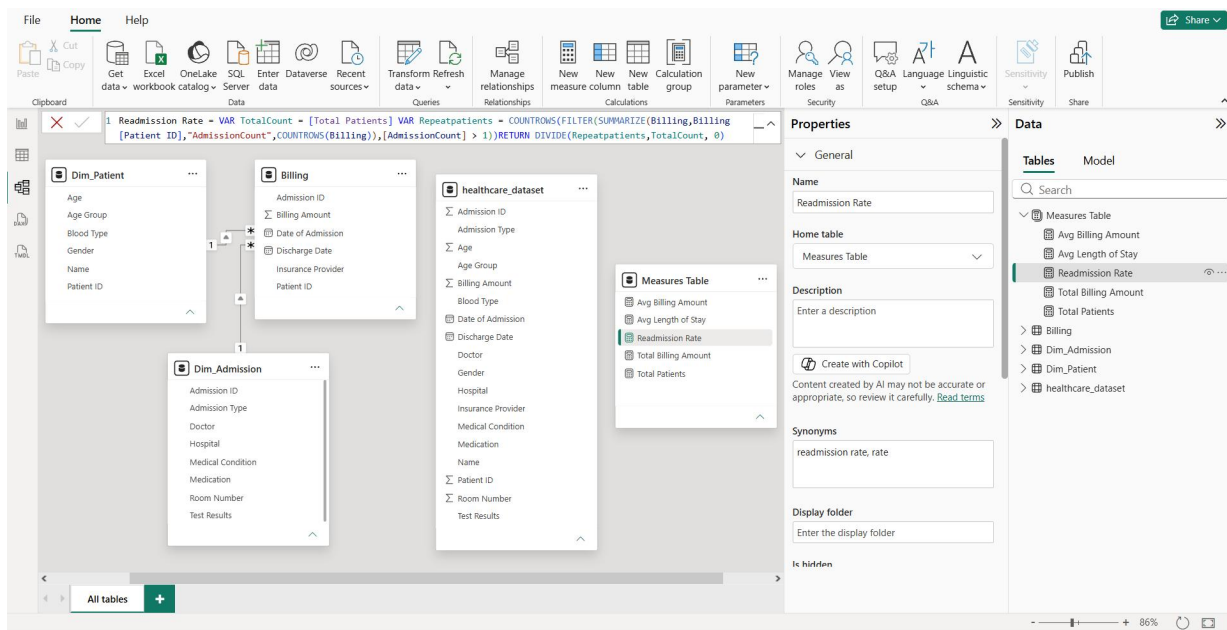
13. Writing the Readmission Rate Measure:

The Readmission Rate measure was written in the formula bar of Power BI Desktop. It reads the Total Patients measure into a variable called TotalCount. It then summarises the Billing table by Patient ID, adds an AdmissionCount column that counts the billing rows per patient, and keeps only the patients whose AdmissionCount is greater than 1. Counting these rows gives Repeatpatients, and the measure returns Repeatpatients divided by TotalCount using DIVIDE, which returns 0 instead of an error if TotalCount is zero.

```
Readmission Rate =
VAR TotalCount = [Total Patients]
VAR Repeatpatients =
    COUNTROWS(
        FILTER(
            SUMMARIZE(
                Billing,
                Billing[Patient ID],
                "AdmissionCount", COUNTROWS(Billing)
            ),
            [AdmissionCount] > 1
        )
    )
RETURN
    DIVIDE(Repeatpatients, TotalCount, 0)
```

In the Properties pane the measure is named Readmission Rate, its Home table is Measures Table, and its synonyms are set to "readmission rate, rate" so it can be found in Q&A. The placeholder

Column1 has also been removed, so the Measures Table now contains only the five measures: Avg Billing Amount, Avg Length of Stay, Readmission Rate, Total Billing Amount, and Total Patients.



14. Conclusion:

The raw healthcare dataset was cleaned and restructured in Power Query. Patient names were standardized, an Age Group conditional column was derived, and Patient ID and Admission ID surrogate keys were added. The data was then split into a Dim_Patient dimension, a Dim_Admission dimension, and a Billing table, and the result was loaded into Power BI Desktop. One-to-many relationships from Dim_Patient and Dim_Admission to Billing complete the star-schema model. A Measures Table was then added with Total Patients, Total Billing Amount, Avg Billing Amount, Avg Length of Stay, and a DAX-based Readmission Rate, so the model is ready for clinical analytics.